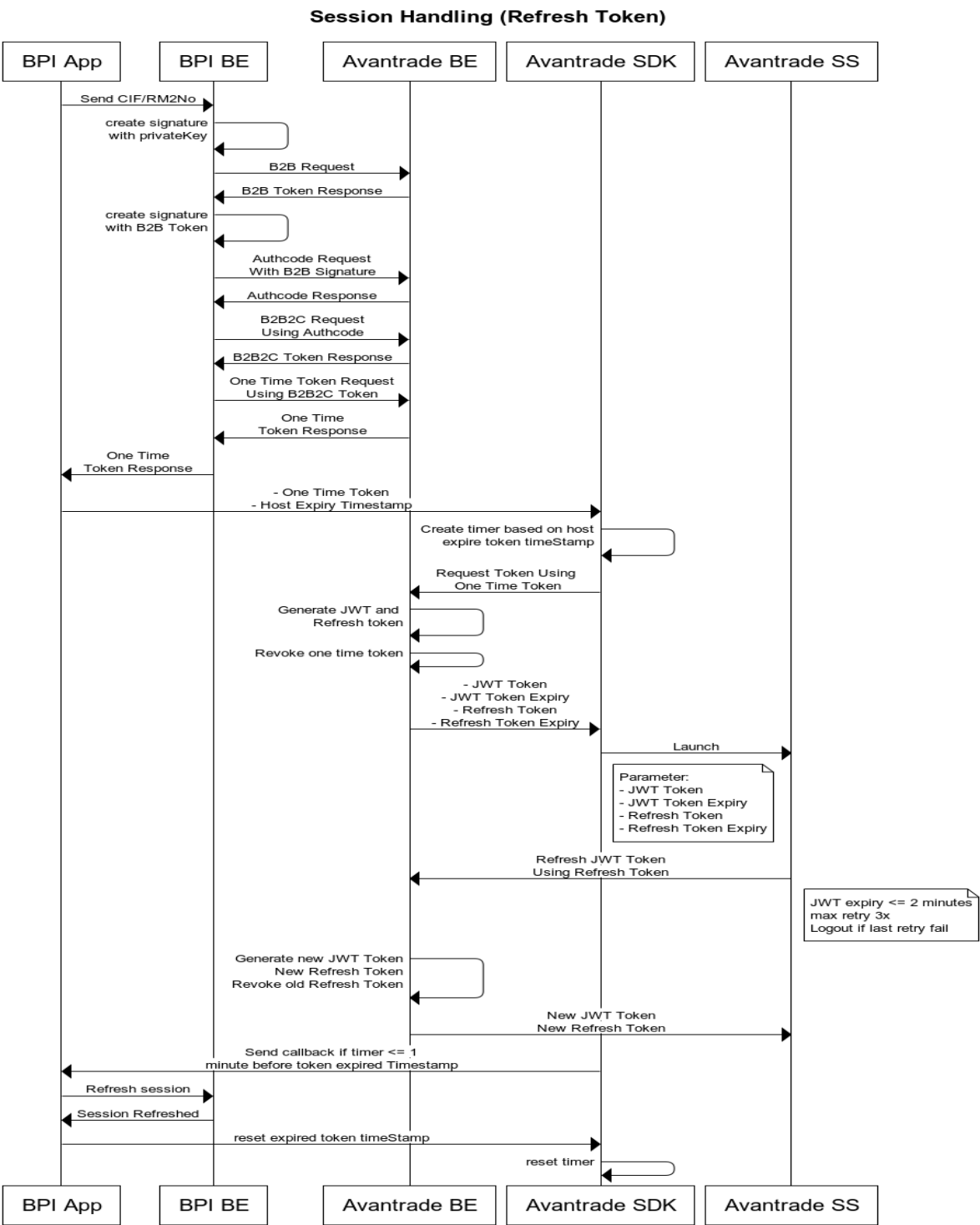


BPI Implementation

1. Session Handling Flow



2. B2B Request Signature

Ensures non-repudiation & integrity checking using asymmetric signature **SHA256withRSA**

(`Private_Key` , `stringToSign`).

- `Private_Key` is provided during the partner registration process

- `stringToSign = X-CLIENT-KEY + "|" + X-TIMESTAMP`

2.1 Function

```
public String generateAuthSignature(
    String clientId,
    String timestamp,
    String privateKeyPem) throws Exception {

    String cleaned = privateKeyPem
        .replace("-----BEGIN PRIVATE KEY-----", "")
        .replace("-----END PRIVATE KEY-----", "")
        .replace("-----BEGIN RSA PRIVATE KEY-----", "")
        .replace("-----END RSA PRIVATE KEY-----", "")
        .replaceAll("\\s", "");

    byte[] keyBytes = Base64.getDecoder().decode(cleaned);
    PrivateKey privateKey = KeyFactory.getInstance("RSA")
        .generatePrivate(new PKCS8EncodedKeySpec(keyBytes));

    String payload = clientId + "|" + timestamp;

    Signature signature = Signature.getInstance("SHA256withRSA");
    signature.initSign(privateKey);
    signature.update(payload.getBytes(StandardCharsets.UTF_8));

    return Base64.getEncoder().encodeToString(signature.sign());
}
```

2.1.1 Function Parameter

Key	Type	Length	Mandatory	Description	Sample Value
clientId	String	N.A	Y	Client UUID from Registration Process	53e75332-b619-4688-ad45-37d279110a16
timestamp	String	N.A	Y	TimeStamp when initiating process	2026-05-20T13:45:12.123+07:00
privateKeyPem	String	N.A	Y	Private Key from Registration Process	-----BEGIN PRIVATE KEY-----
					asdasgdjhg2jg21hj3g1jg3jh1g23hj12g
					-----END PRIVATE KEY-----

2.2 Usage

```
ZonedDateTime now = ZonedDateTime.now(TimeZone.getTimeZone("Asia/Jakarta").toZoneId());
String timeStamp = now.format(DateTimeFormatter.ofPattern("yyyy-MM-dd'T'HH:mm:ss.SSSXXX"));
String privateKeySignature;
try {
    privateKeySignature = helper.generateAuthSignature(
        constants.getClientId2(),
```

```
        timeStamp,  
        constants.getPrivateKeyPem2());  
} catch (Exception e) {  
    throw new CustomException(e.toString());  
}
```

3. B2B2C Request Signature

Represents the signature of a request using symmetric signature **HMAC_SHA512** (`secretKey` , `stringToSign`).

```
stringToSign = HTTPMethod + ":" + EndpointURL + ":" + access_token + ":" +  
Lowercase(HexEncode(SHA-256(requestBody))) + ":" + X-TIMESTAMP
```

If there is no request body, use an empty string `""`.

3.1 Function

```
public static String generateTransactionSignature(  
    String httpMethod,  
    String endPoint,  
    String token,  
    String timestamp,  
    String body,  
    String clientSecret) throws NoSuchAlgorithmException, InvalidKeyException {  
  
    String hexEncodeBody = sha256Hex(body);  
    String stringToSign = String.join(":", httpMethod, endPoint, token, hexEncodeBody, timestamp);  
  
    SecretKeySpec secretKey = new SecretKeySpec(clientSecret.getBytes(), "HmacSHA512");  
    Mac hmac = Mac.getInstance("HmacSHA512");  
    hmac.init(secretKey);  
  
    return Base64.getEncoder().encodeToString(  
        hmac.doFinal(stringToSign.getBytes(StandardCharsets.UTF_8)));  
}
```

3.1.1 Function Parameter

Key	Type	Length	Mandatory	Description	Sample Value
httpMethod	String	N.A	Y	endpoint method	POST
endPoint	String	N.A	Y	url endpoint	/auth-service/authentication/v2/access-token/b2b
token	String	N.A	Y	B2B Token	asdasgdjhg2jg21hj3g1jg3jh1g23hj12g
timestamp	String	N.A	Y	TimeStamp when initiating process	2026-05-20T13:45:12.123+07:00
body	String	N.A	Y	Empty String by default	
clientSecret	String	N.A	Y	channel id from B2B Token Response	53e75332-b619-4688-ad45-37d279110a16

3.2 Usage

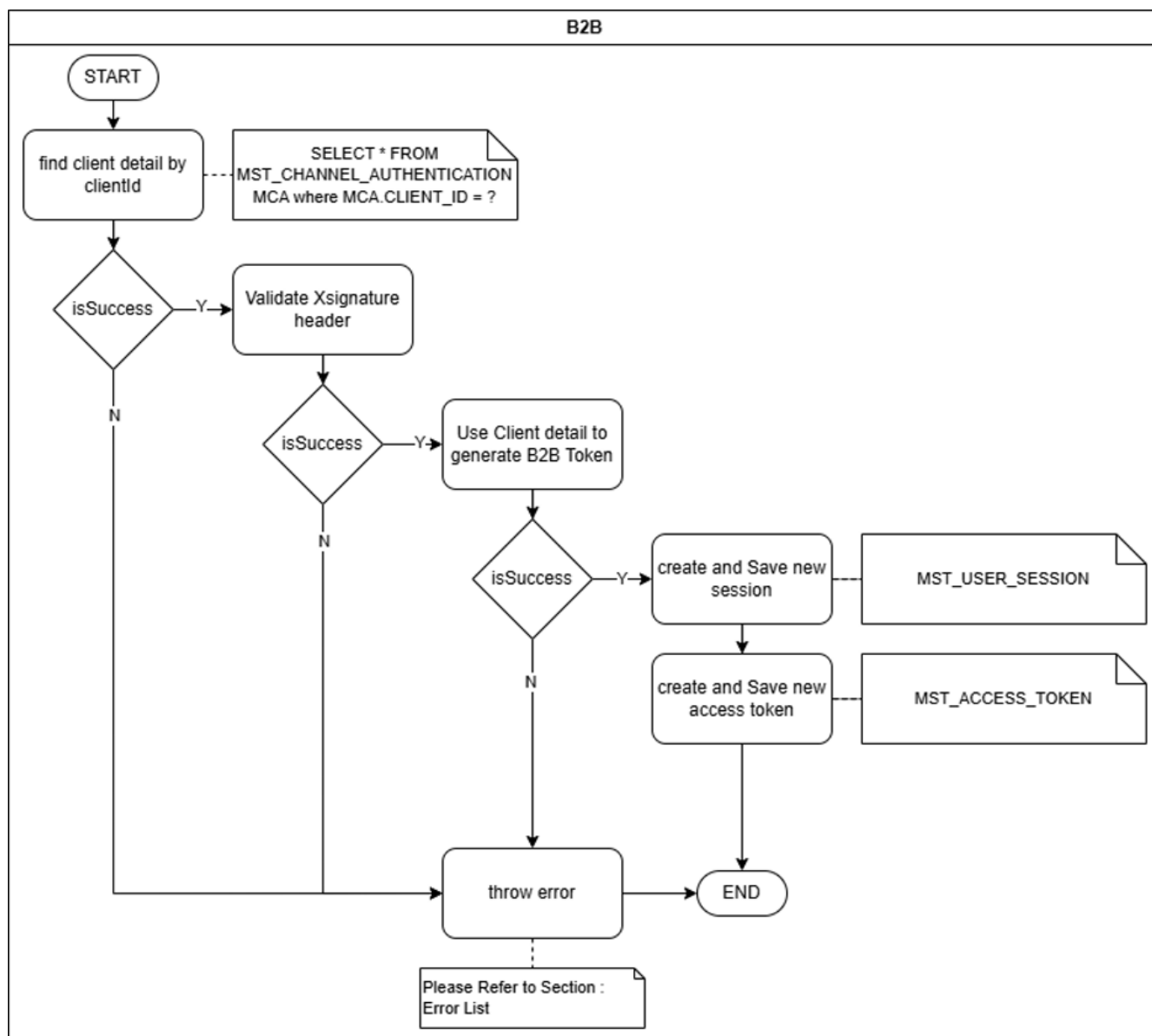
```
public String generateTransactionSignature(String timeStamp, UUID channelId, String b2bToken) {  
    try {
```

```
        return SignatureUtil.generateTransactionSignature(
            "POST",
            "/auth-service/authentication/v2/access-token/b2b",
            b2bToken,
            timeStamp,
            "",
            channelId.toString());
    } catch (Exception e) {
        throw new RuntimeException(e);
    }
}
```

4. API Detail

4.1 B2B

4.1.1 Flowchart



4.1.2 URL

POST auth-service/authentication/v2/access-token/b2b

4.1.3 Header Parameter

Key	Type	Length	Mandatory	Description	Sample Value
X-CLIENT-KEY	String	N.A	Y	clientId from registration process	c04ee779-1a81-4453-83ea-fd370e9f4cc9
X-TIMESTAMP	String	N.A	Y	Timestamp when create secretKey and privateKey signature	7xDTUTKUIUF1e5XC9OsBxNJEN4C...
X-SIGNATURE	String	N.A	Y	privateKey Signature	SIGNATURE

4.1.4 Body Parameter

Key	Type	Length	Mandatory	Description	Sample Value
grantType	String	N A	Y	-	client_credentials

4.1.5 Response

Property	Type Data	Sample Value
accessToken	String	8S1NjMEMF3lqQr2Q1PgPAN26j1aA
tokenType	String	Bearer
expiresIn	String	900 (Seconds)
sessionId	UUID	d568e0f0-740e-42fc-a65e-c8860a50cd3d
channelId	UUID	d568e0f0-740e-42fc-a65e-c8860a50cd3d
successCode	String	43
successMessage	String	Success

4.1.6 Data Relations

Property	Mapping	Relations
accessToken	OAuth2Token.getTokenValue	
tokenType	OAuth2AccessToken.getTokenType	
sessionId	MST_USER_SESSIONS.SESSION_ID	

4.1.7 Curl Example

```
curl --location '<host>/api-gateway/auth-service/authentication/v2/access-token/b2b' \
--header 'X-CLIENT-KEY: 47c5963a-4836-d939-e063-e30a1aac30ba' \
--header 'X-TIMESTAMP: 2025-07-11T14:21:23.456+08:00' \
--header 'X-SIGNATURE: cX5cBkPQwaXaV8qbRcwc/EmqI9spIEKrBQDCllsHgDD/AgqjS9Y9DQxkJX1UMHdy5z9L4YgrqSLvImD2HxwPfv2wAx0HBd+HENSx6TgX/YCzFex0jqUk12FML4B8wK13Eqp9ARbkrGAKhnlcUW+Q3IcTFM+miq+SUq0Q8j3vQjbBSKdpWgS2Qn0bPAu0BCXqmQaYkKZ7AP03bA/VKmkXi0ixq2H7/U8Y0qeX0K3hznYZnPZCMIkhJfamLXBHqDdLHUtiYoL3WaYTj+q+vw4Uah4Hj3I1zEUrkphR3lh10o3pF7EJUVqGYBrEYqJRKgBlT15T9z5UvEMbonGyCDK8Hw==' \
--header 'Content-Type: application/json' \
--data '{
```

```
}'  "grantType":"client_credentials"
```

4.1.8 Response Example

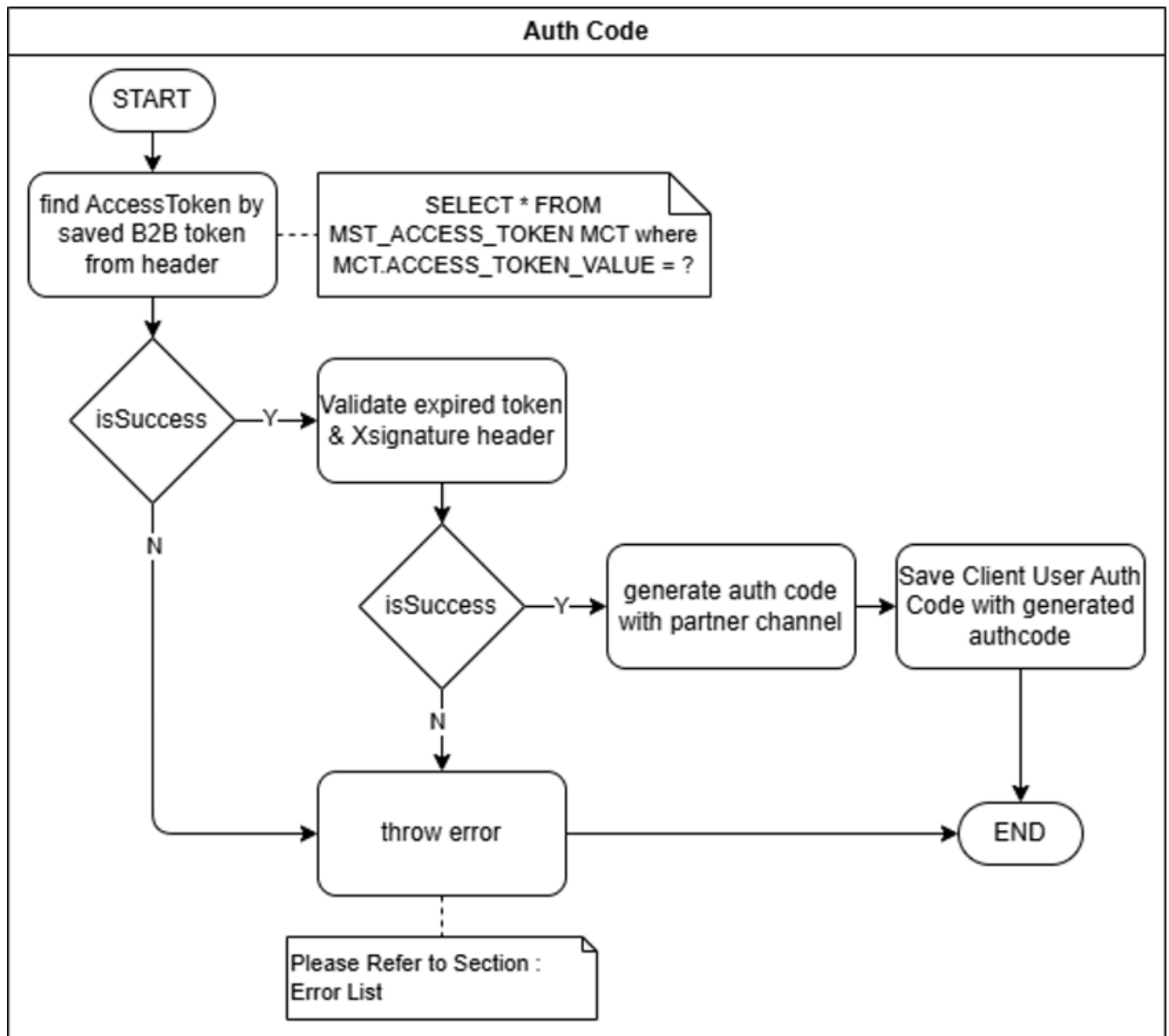
```
{  "accessToken": "eyJhbGciOiJIUzI1NiJ9.dummy.payload.signature",  "tokenType": "Bearer",  "expiresIn": "900",  "sessionId": "550e8400-e29b-41d4-a716-446655440000",  "channelId": "WEB",  "additionalInfo": {}}
```

4.1.9 Error List

Code	Message Error	HTTP Code
AUTH_00	Internal Server Error	500
AUTH_01	Invalid Login User!	400

4.2 Auth Code

4.2.1 Flowchart



4.2.2 URL

POST auth-service/authentication/v2/generate-auth-code/b2b2c

4.2.3 Header Parameter

Key	Type	Length	Mandatory	Description	Sample Value
X-PARTNER-ID	String	N.A	Y	clientId	c04ee779-1a81-4453-83ea-fd370e9f4cc9
X-TIMESTAMP	String	N.A	Y		7xDTUTKUIUF1e5XC9OsBxNJEN4C...

Key	Type	Length	Mandatory	Description	Sample Value
				Timestamp when create secretKey signature	
X-SIGNATURE	String	N.A	Y	secretKey Signature	SIGNATURE
X-BRANCH-ID	String	N.A	Y	clientId	c04ee779-1a81-4453-83ea-fd370e9f4cc9
X-SESSION-ID	String	N.A	Y	Session id from B2B response	5207d4a3-5666-4d6d-8f1f-7732ca8fa85e
Authorization	String	N.A	Y	B2B token	7xDTUTKUIUF1e5XC9OsBxNJEN4C...

4.2.4 Body Parameter

Key	Type	Length	Mandatory	Description	Sample Value
paramType	String	N.A	Y	param type used	CIF_BANK
paramValue	String	N.A	Y	param value used for get customer data	CIF2501045138
partnerChannel	String	N.A	Y	registered partner channel	CH_BPI

4.2.5 Response

Property	Type Data	Sample Value
authCode	String	8S1NjMEMF3lqQr2Q1PgPAN26j1aA
successCode	String	43
successMessage	String	Success

4.2.6 Data Relations

Property	Mapping	Relations
authCode	MST_CLIENT_USER_AUTH_CODE.AUTH_CODE	

4.2.7 Curl Example

```
curl --location '<host>/api-gateway/auth-service/authentication/v2/generate-auth-code/b2b2c' \
--header 'Authorization: ato3DmdyH0nnnCKLB9jInGdmzMfPreV6BwUs8-6bYMBGg81Wx2ayEqHI1DRD7BRghPAhvS10eYnRW7bXL20_U4BCer7yZSRb0blq5iTMux0XKIZ-1u8rI1IX9uYUEZEH' \
--header 'X-PARTNER-ID: 47c5963a-4836-d939-e063-e30a1aac30ba' \
--header 'X-BRANCH-ID: 47c5963a-4836-d939-e063-e30a1aac30ba' \
--header 'X-TIMESTAMP: 2025-07-11T14:21:23.456+08:00' \
--header 'X-SIGNATURE: Ymk6zVb0Cjld9Jd7MdVYyD3DgZPn1pssFr8KgXB/d0z9Tmue6ul5UmpvBDqBKdKD0Jk9mKh0smBV4LiraJYvQ==' \
--header 'Content-Type: application/json' \
--data '{
  "paramType": "CIF_BANK",
  "paramValue": "00000001102859",
  "partnerChannel": "CH_BPI"
}'
```

4.2.8 Response Example

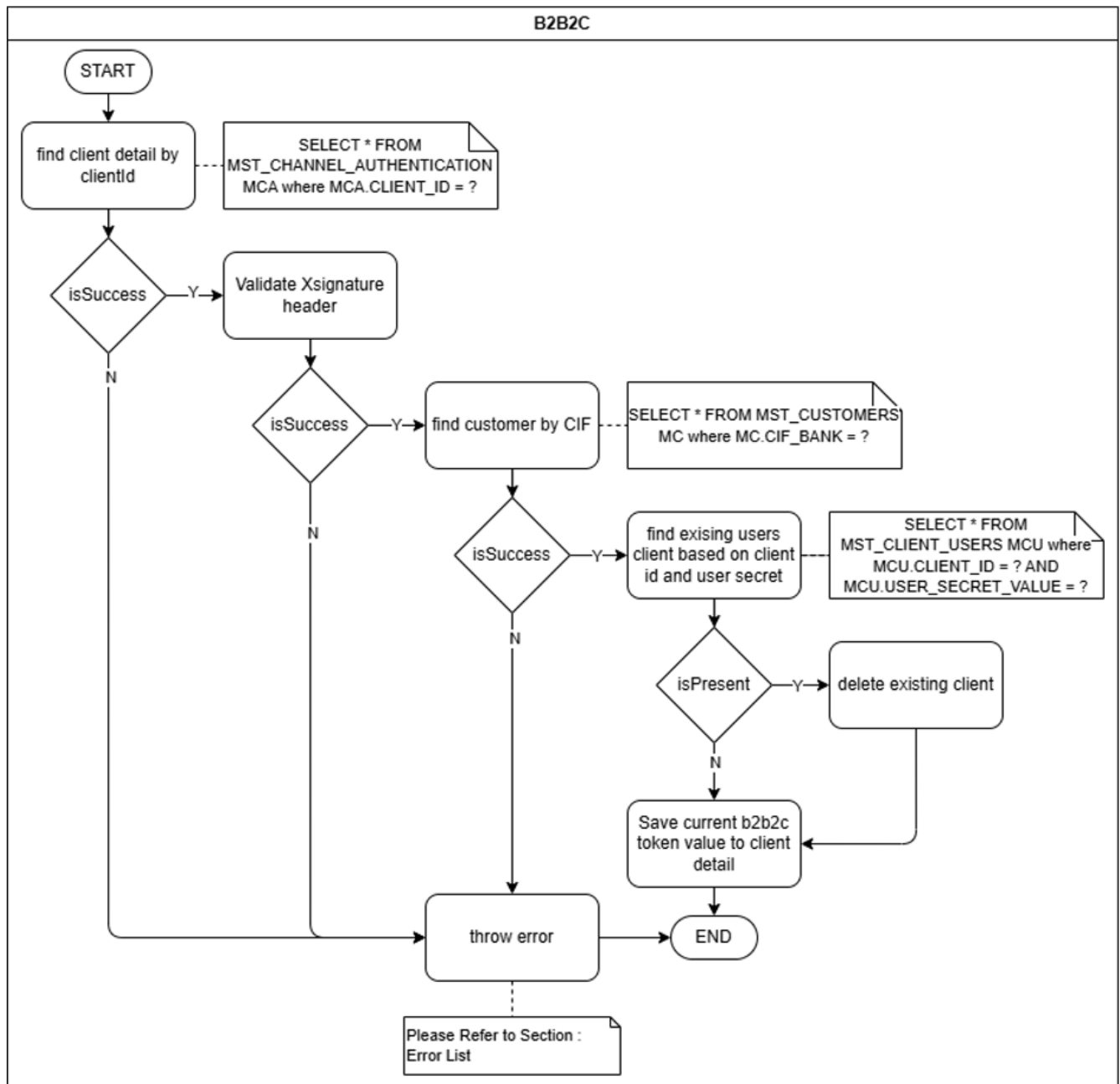
```
{
  "successCode": "43",
  "successMessage": "Success",
  "authCode": "AUTH-7F3A9C8D5B2E1F4A"
}
```

4.2.9 Error List

Code	Message Error	HTTP Code
AUTH_00	Internal Server Error	500
AUTH_01	Invalid Login User!	400

4.3 B2B2C

4.3.1 Flowchart



4.3.2 URL

POST auth-service/authentication/v2/access-token/b2b2c

4.3.3 Header Parameter

Key	Type	Length	Mandatory	Description	Sample Value
X-CLIENT-KEY	String	N.A	Y	clientId	

Key	Type	Length	Mandatory	Description	Sample Value
					c04ee779-1a81-4453-83ea-fd370e9f4cc9
X-TIMESTAMP	String	N.A	Y	Timestamp when create secretKey and privateKey signature	7xDTUTKUIUF1e5XC9OsBxNJEN4C...
X-SIGNATURE	String	N.A	Y	privateKey Signature	SIGNATURE
X-BRANCH-ID	String	N.A	Y	clientId	c04ee779-1a81-4453-83ea-fd370e9f4cc9
X-SESSION-ID	String	N.A	Y	Session id from B2B response	5207d4a3-5666-4d6d-8f1f-7732ca8fa85e
AUTHORIZATION-CUSTOMER	String	N.A	Y	token from B2B Response	5207d4a3-5666-4d6d-8f1f-7732ca8fa85e

4.3.4 Body Parameter

Key	Type	Length	Mandatory	Description	Sample Value
grantType	String	N.A	Y	grant type	AUTHORIZATION_CODE
authCode	String	N.A	Y	auth code from response	eyJhbGciOiJIUzUxMij9
refreshToken	String	N.A	Y	empty by default	-

4.3.5 Response

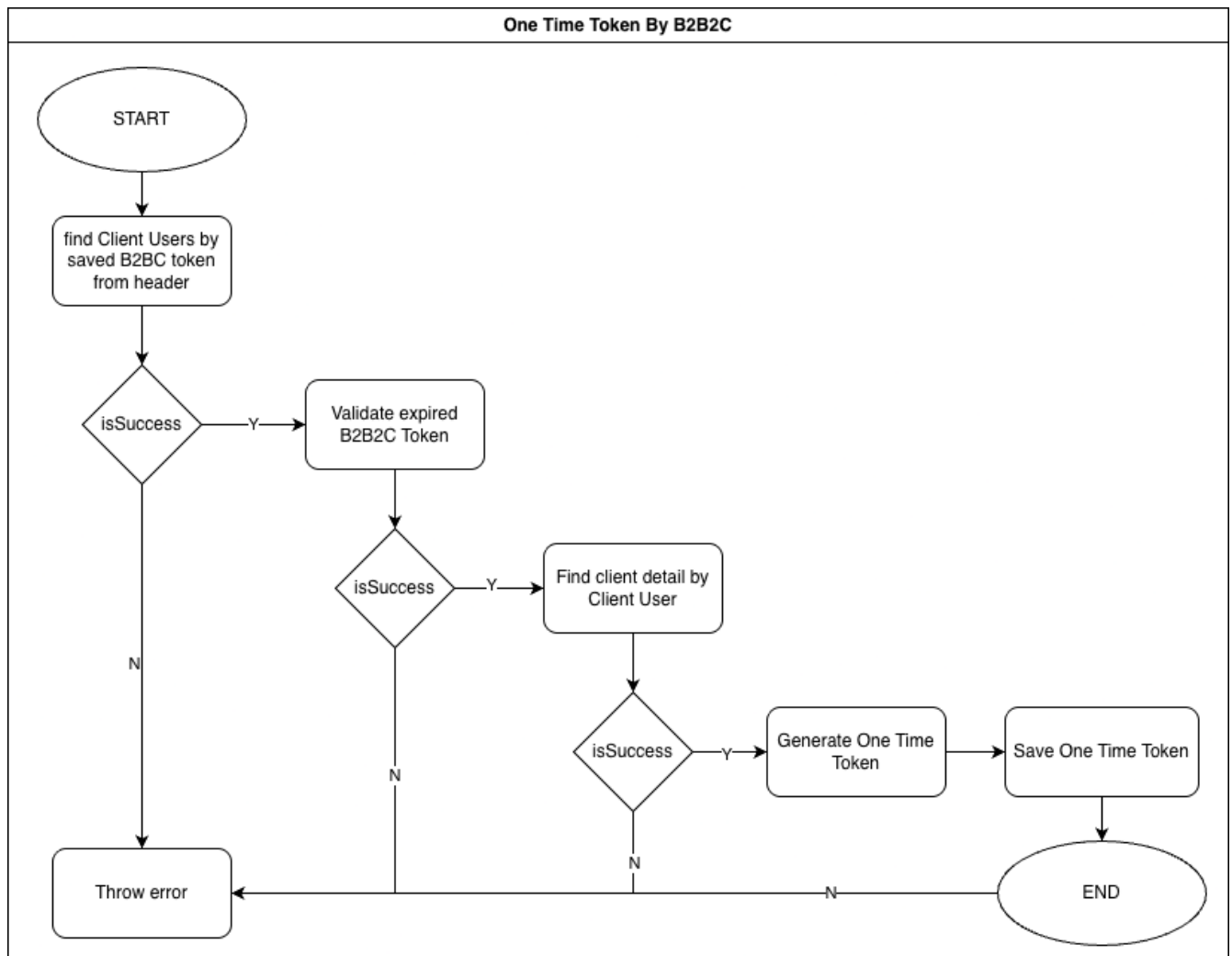
Property	Type Data	Sample Value
accessToken	String	B2B2C Token
tokenType	String	Bearer
accessTokenExpiryTime	String	1296000 (In Seconds - 15 Days)
refreshToken	String	8S1NjMEMF3lqQr2Q1PgPAN26j1aA
refreshTokenExpiryTime	String	900
responseCode	String	43
responseMessage	String	Success

4.3.6 Data Relations

Property	Mapping	Relations
accessToken	OAuth2Token.getTokenValue	
tokenType	OAuth2AccessToken.getTokenType	
accessTokenExpiryTime		

4.4 One Time Token By B2B2C

4.4.1 Flowchart



4.4.2 URL

POST auth-service/authentication/v2/token/by-b2bc-token

4.4.3 Header Parameter

Key	Type	Length	Mandatory	Description	Sample Value
authorization-customer	String	N.A	Y	B2B2C Token	7SZjVjOzWfm_VuPqilbieMAG8sh7UxhicaCz46h-GqmgTTIW5NXpxCS5dBCjuFw_AODJGoBg9XgFHY_IG5FkLzbYa
Authorization	String	N.A	Y	B2B2C Token	7SZjVjOzWfm_VuPqilbieMAG8sh7UxhicaCz46h-GqmgTTIW5NXpxCS5dBCjuFw_AODJGoBg9XgFHY_IG5FkLzbYa

4.4.4 Response

Property	Type Data	Sample Value
successCode	String	43
successMessage	String	Success
oneTimeToken	String	eyJhbGciOiJIUz...
oneTimeTokenExpiry	String	900

4.4.5 Data Relations

Property	Mapping	Relations
oneTimeToken	MST_ONE_TIME_TOKEN.TOKEN	
oneTimeTokenExpiry	MST_ONE_TIME_TOKEN.EXPIRED_AT	

4.4.6 Curl Example

```
curl -X POST <host>/api-gateway/auth-service/authentication/v2/token/by-b2bc-token \
-H "authorization-customer: <B2B2C_ACCESS_TOKEN>" \
-H "Authorization: <B2B2C_ACCESS_TOKEN>" \
-H "ENV: SIT"
```

4.4.7 Response Example

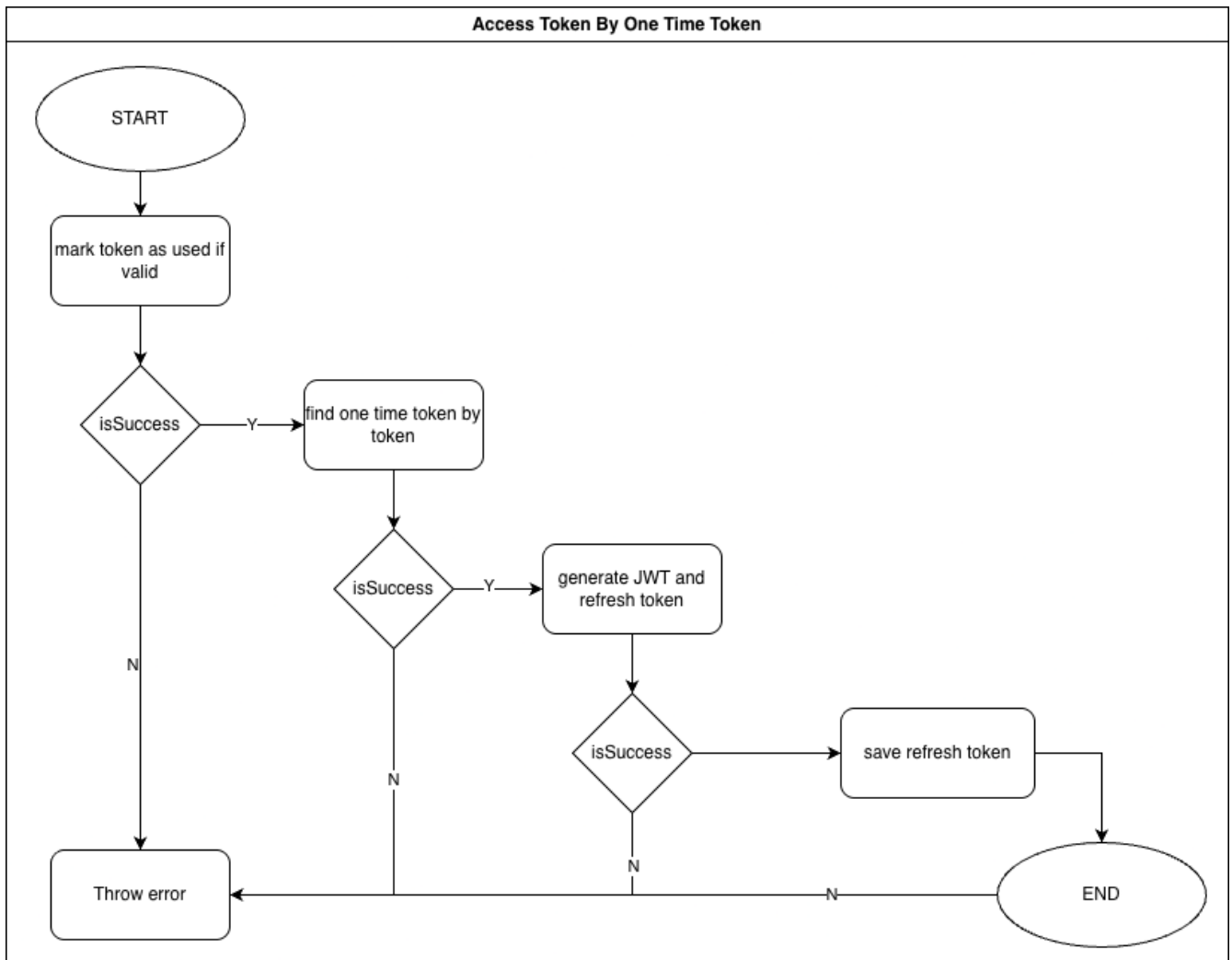
```
{
  "successCode": "43",
  "successMessage": "Success",
  "oneTimeToken": "eyJhbGciOiJIUzI1NiJ9..",
  "oneTimeTokenExpiry": "2026-06-04T09:35:49+07:00"
}
```

4.4.8 Error List

Code	Message Error	HTTP Code
AUTH_00	Internal Server Error	500
AUTH_01	Invalid Login User!	400

4.5 Access Token By One Time Token

4.5.1 Flowchart



4.5.2 URL

POST auth-service/authentication/v2/token/by-one-time-token

4.5.3 Header Parameter

Key	Type	Length	Mandatory	Description	Sample Value
ENV	String	N.A	Y	Selected Environment (DEV/GAD/SIT/UAT)	DEV

4.5.4 Body Parameter

Key	Type	Length	Mandatory	Description	Sample Value
oneTimeToken	String	N.A	Y	One Time Token	eyJhbGciOiJIUzUxMiJ9...

4.5.5 Response

Property	Type Data	Sample Value
successCode	String	43
successMessage	String	Success
jwtToken	String	eyJhbGciOiJIUzUxMij9...
jwtTokenExpiry	String	900
refreshToken	String	eyJhbGciOiJIUzUxMij9...
refreshTokenExpiry	String	TimeStamp
refreshTokenExpiry	String	TimeStamp
webViewUrl	String	dynamic Webview URL

4.5.6 Data Relations

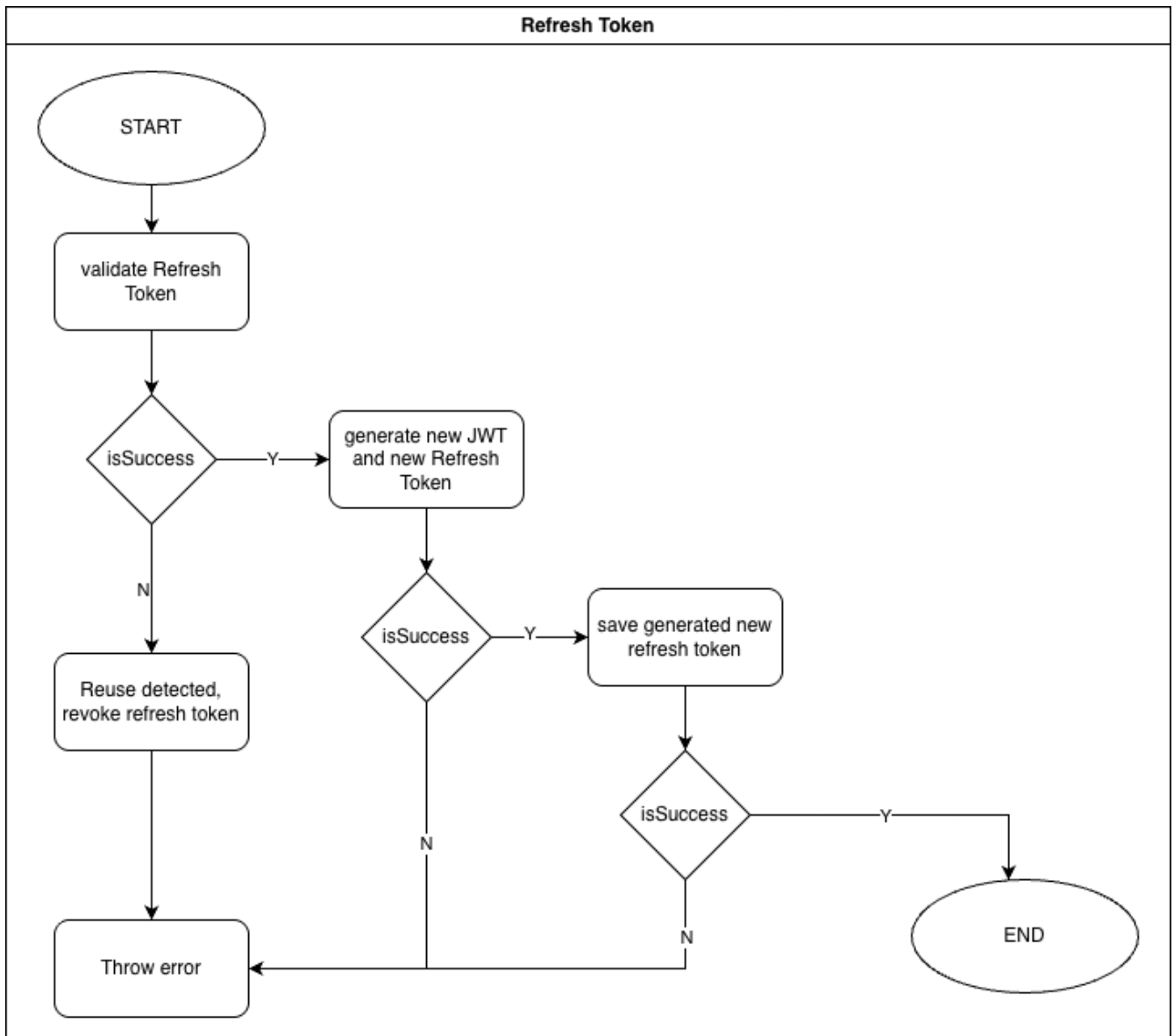
Property	Mapping	Relations
refreshTokenExpiry	MST_REFRESH_TOKEN.EXPIRED_AT	

4.5.7 Error List

Code	Message Error	HTTP Code
AUTH_00	Internal Server Error	500
AUTH_01	Invalid Login User!	400

4.6 Refresh Token

4.6.1 Flowchart



4.6.2 URL

POST auth-service/authentication/v2/refresh-token

4.6.3 Body Parameter

Key	Type	Length	Mandatory	Description	Sample Value
jwtToken	String	N.A	Y	Old JWT Token	eyJhbGciOiJIUzUxMiJ9...
refreshToken	String	N.A	Y	Current Refresh Token	eyJhbGciOiJIUzUxMiJ9...

4.6.4 Response

Property	Type Data	Sample Value
successCode	String	43

Property	Type Data	Sample Value
successMessage	String	Success
jwtToken	String	eyJhbGciOiJIUzUxMiJ9...
jwtTokenExpiry	String	900
refreshToken	String	eyJhbGciOiJIUzUxMiJ9...
refreshTokenExpiry	String	TimeStamp

4.6.5 Data Relations

Property	Mapping	Relations
refreshTokenExpiry	MST_REFRESH_TOKEN.EXPIRED_AT	

4.6.6 Error List

Code	Message Error	HTTP Code
AUTH_04	Invalid or expired refresh token	400
AUTH_00	Internal Server Error	500